

MYS BYKOV, Russian Federation

WMO: 218210

Lat: 72.004N

Lon: 129.118E

Elev: 16

StdP: 101.13

Time Zone: 9.00 (E09)

Period: 84-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.2	-38.7	-43.2	0.1	-39.7	-41.8	0.1	-38.5	17.1	-23.3	15.7	-23.8	1.9	140	1.076

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	17.3	14.0	15.1	12.6	13.3	11.2	14.2	16.8	12.7	14.8	11.4	13.3	4.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.8	9.2	15.7	11.6	8.5	14.4	10.0	7.6	12.9	40.2	17.1	36.3	14.9	32.7	13.4	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.7	12.9	11.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-41.7	15.9	2.6	1.8	-43.6	17.2	-45.2	18.2	-46.6	19.2	-48.5	20.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.2	-30.2	-29.5	-23.4	-14.9	-7.1	3.6	8.3	8.6	3.1	-8.2	-19.6	-26.4
	DBStd	15.00	4.77	6.06	8.27	6.65	4.56	4.32	3.32	3.13	3.41	5.38	5.02	5.18
	HDD10.0	7791	1247	1106	1036	748	531	197	74	66	208	564	887	1127
	HDD18.3	10782	1505	1339	1294	998	789	441	311	302	457	823	1137	1386
	CDD10.0	51	0	0	0	0	0	7	22	22	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.1	4.5	4.5	4.7	4.3	4.7	5.6	5.8	5.6	5.4	5.5	5.4	4.7
Precipitation	PrecAvg	397	27	19	20	15	29	37	54	57	42	38	27	32
	PrecMax	569	74	66	52	70	72	71	121	126	105	66	47	97
	PrecMin	273	4	1	4	0	5	11	18	27	15	12	10	2
	PrecStd	72	17	16	12	14	16	15	25	22	20	15	11	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-16.9	-12.6	-2.9	0.5	4.7	18.9	20.4	19.5	12.6	1.9	-5.5	-13.3
		MCWB	-17.4	-13.1	-4.0	-0.9	2.4	14.5	15.9	14.5	10.1	1.1	-5.8	-13.9
	2%	DB	-19.6	-15.9	-5.3	-1.7	2.1	14.8	17.9	16.6	10.5	0.4	-8.3	-15.3
		MCWB	-19.9	-16.2	-6.0	-2.8	0.9	11.6	14.7	13.5	8.5	-0.4	-8.8	-15.8
	5%	DB	-21.7	-19.2	-7.7	-3.6	1.0	12.1	15.6	14.7	9.2	-0.8	-10.4	-17.2
		MCWB	-22.0	-19.6	-8.4	-4.5	0.1	9.9	13.0	12.4	7.8	-1.6	-10.8	-17.6
	10%	DB	-23.4	-21.3	-11.1	-5.8	-0.1	9.5	13.5	13.2	7.9	-1.7	-12.7	-19.2
		MCWB	-23.7	-21.6	-11.6	-6.6	-0.7	7.7	11.5	11.2	6.5	-2.5	-13.1	-19.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-17.3	-13.0	-3.6	-0.8	2.8	14.7	16.9	15.3	10.3	0.7	-5.7	-13.7
		MCDB	-17.0	-12.7	-3.0	0.5	4.6	18.5	19.3	18.4	11.9	1.3	-5.5	-13.4
	2%	WB	-19.9	-16.3	-5.9	-2.8	1.1	11.8	14.9	13.6	9.0	-0.4	-8.7	-15.7
		MCDB	-19.7	-15.9	-5.3	-1.8	2.0	14.6	17.6	15.9	10.0	0.3	-8.3	-15.4
	5%	WB	-22.0	-19.5	-8.3	-4.5	0.2	9.9	13.3	12.6	7.8	-1.5	-10.9	-17.6
		MCDB	-21.8	-19.2	-7.7	-3.6	0.9	12.0	15.4	14.5	9.1	-0.8	-10.6	-17.2
	10%	WB	-23.7	-21.7	-11.6	-6.6	-0.7	7.7	11.7	11.4	6.6	-2.5	-13.0	-19.5
		MCDB	-23.5	-21.4	-11.2	-5.8	0.0	9.3	13.6	13.2	7.9	-1.8	-12.7	-19.3
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.8	5.7	7.1	5.1	4.7	4.3	4.0	2.9	3.0	4.6	4.4
		MCDBR	6.0	7.2	6.5	9.2	6.6	9.0	7.7	7.2	5.0	2.6	5.7	6.2
		MCWBR	5.8	7.0	6.5	8.5	5.9	6.6	5.6	5.5	4.0	2.5	5.6	6.1
	5% WB	MCDBR	5.8	7.1	6.8	9.0	6.1	8.8	7.5	6.8	4.8	2.5	5.8	6.2
		MCWBR	5.8	7.0	6.6	8.4	5.5	6.6	5.7	5.6	4.1	2.4	5.7	6.1
Clear-Sky Solar Irradiance	taub	N/A	0.194	0.277	0.306	0.336	0.339	0.395	0.375	0.300	0.206	N/A	N/A	
	taud	N/A	1.944	2.017	2.052	2.090	2.311	2.278	2.339	2.448	2.015	N/A	N/A	
	Ebn at Noon	0	439	666	794	820	840	763	728	690	397	0	0	
	Edh at Noon	0	32	78	110	123	103	104	86	54	28	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.22	1.68	4.13	5.91	6.04	4.58	2.81	1.36	0.37	0.00	0.00	
	RadStd	0.00	0.02	0.08	0.16	0.25	0.41	0.39	0.33	0.14	0.06	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+1.05	N/A	N/A	N/A	N/A	N/A	-413	-394	N/A	N/A

Nomenclature: See separate page